Zhe Yu

Research Assistant | Trustworthy Language Models and Agentic AI
B.Eng. Candidate in Artificial Intelligence | Expected 2027
HKUST (Guangzhou) | LAI Lab · Zhejiang University | IFRC Lab
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Research Profile

Undergraduate researcher developing white-box and system-level methods for trustworthy language models and agents, with a focus on evidence attribution, representation–action dissociation, malicious skill detection, and multi-agent execution consistency.

Education

Communication University of Zhejiang **2023 – Expected 2027**
B.Eng. in Artificial Intelligence; advisor: Dr. Hao Zeng Hangzhou, China
Overall Average: 85.44/100

Research Experience

The Hong Kong University of Science and Technology (Guangzhou) **May 2026 – Present**
Research Assistant, LAI Lab; advised by Prof. Chengwei Qin Guangzhou, China

- Developed CAVERT for diagnosing and recovering from execution inconsistencies in multi-agent systems; formalized task duties and evidence conditions that distinguish corrected memory from compliant execution.
- Co-authored a survey of hallucinations across the reasoning lifecycle, connecting observable evidence and causal study designs to verification, repair, and release control.

George Mason University (Remote) **Jun. – Sep. 2026**
Research Intern; supervised by Prof. Simin Chen Fairfax, VA, USA

- Developed LatentTriad, a pre-action detector that probes declaration, operation, and boundary representations in a single prefill pass to identify malicious agent skills without executing them; studied robustness across skill sources and lexical decoys.

Binjiang Institute of Zhejiang University **Nov. 2025 – Present**
Research Intern, IFRC Lab; supervised by Prof. Meng Han and Dr. Wenpeng Xing Hangzhou, China

- Developed DISF and FIDES for white-box RAG reliability, and Composition Collapse for distinguishing stable factual knowledge from reliable compositional reasoning. FIDES and Composition Collapse were accepted at EMNLP 2026.
- Developed evidence-attribution, audit-planning, and evaluator-reliability methods; studied representation–action dissociation in Readable Before Actionable and Decodable but Unresolved.
- Within provincial and national key R&D programs, developed LatentAudit and RETINA-SAFE / ECRT for faithfulness monitoring and medical risk triage; both accepted at NeurIPS 2026 workshops. Contributed to privacy-preserving model ownership verification through ZK-FPE.

University of Malaya **Jan. – Feb. 2025**
Visiting Student Kuala Lumpur, Malaysia

- Contributed to a medical computer-vision project in collaboration with the medical school at the National University of Singapore.

Westlake University **Mar. – Sep. 2024**
Research Assistant; supervised by Dr. Ziyang Zhang Hangzhou, China

- Worked on parallel dual all-fiber interferometers for salinity and temperature sensing; the study was published in *Optics Communications* (2025).

Industry Experience

BoostEngine **Jul. – Oct. 2025**
R&D Intern, Full-Stack Development / AI Agents Hangzhou, China; remote with New York team

- Built a multi-brand Manifest V3 Chrome extension for TikTok Shop operations across 20 regional domains and 9 language packs using React, TypeScript, Vite, and Chrome APIs.
- Developed analytics services for 8,000+ creator collaborations and a FastAPI / Redis / MySQL synchronization system with anti-loop control, queue coalescing, monitoring, and alerts.

Academic Service

Reviewer: EMNLP 2026, NeurIPS 2026, AAAI 2027, ICLR 2027.

Awards and Technical Skills

- **Silver Award**, CCB Cup Zhejiang Provincial International College Students' Innovation Competition, 2024; project: *Scoliosis Detection AI System*.
- **Research:** Python, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, FAISS, vLLM, W&B, MATLAB.
- **Engineering:** TypeScript / JavaScript, Java, React, FastAPI, Spring Boot, SQLAlchemy, MyBatis-Plus, MySQL, Redis, Docker, Chrome Extensions.

Peer-Reviewed Publications

1. **Zhe Yu***, Wenpeng Xing*, Wenjie Luo, Weize Xu, Lingtong Huang, Yourong Chen, Changting Lin, Meng Han. *DISF: Detecting Hallucinations in Retrieval-Augmented Generation via Dual-path Internal State Forcing Framework*. Accepted at **ACL 2026**. [PDF]
2. **Zhe Yu***, Wenpeng Xing*, Tiancheng Zhao, Mohan Li, Changting Lin, Meng Han. *FIDES: Faithful Inference via Deep Evidence Signals for Retrieval-Memory Conflict in RAG*. **EMNLP 2026 Main Conference**. [PDF]
3. **Zhe Yu***, Wenpeng Xing*, Yunzhao Wei, Hongzhi Wang, Xuyang Teng, Meng Han. *Composition Collapse: Stable Factual Knowledge Does Not Imply Compositional Reasoning*. **Findings of EMNLP 2026**. [arXiv]
4. **Zhe Yu***, Wenpeng Xing*, Meng Han. *LatentAudit: Real-Time White-Box Faithfulness Monitoring for Retrieval-Augmented Generation with Verifiable Deployment*. Accepted at **NeurIPS 2026 Workshop**. [PDF]
5. **Zhe Yu***, Wenpeng Xing*, Meng Han. *From Retinal Evidence to Safe Decisions: RETINA-SAFE and ECRT for Hallucination Risk Triage in Medical LLMs*. Accepted at **NeurIPS 2026 Workshop**. [PDF]
6. Zhiguo Ma*, **Zhe Yu**, Wenpeng Xing, Yourong Chen, Meng Han. *ZK-FPE: Blockchain-Verifiable Model Fingerprinting with Zero-Knowledge Privacy for Ownership Attribution*. **ACM TURC 2026**. [PDF]
7. Weiping Yu*, Weihan Wang*, Mingyuan Yan, Keyang He, **Zhe Yu**, Wenpeng Xing, Liyuan Liu, Meng Han. *Scalable and Trusted Metadata-Coordinated Tiered Off-Chain Storage with Dynamic On-Chain Mapping for Recovery-Safe and Low-Latency IoT Data Management*. *Electronics*, 2026, 15(13), 2806. [Publisher]
8. F. Zhou, C. Chang, Q. Chang, H. Zhang, **Zhe Yu**, W. Liu, J. Li, J. Yang. *Orthogonal salinity and temperature detection via paralleled dual all-fiber interferometers*. *Optics Communications*, 583 (2025): 131688. [DOI]
9. **Zhe Yu**, H. Zeng, Y. Zhao, X. Zhang, Z. Wang, Y. Tao, M. Yuan, X. Sun. *Bibliometric analysis of physical education research in China from 2014 to 2024*. **ACM ICETM**, 2025, pp. 128–132. [DOI]

Manuscripts Under Review

1. **Zhe Yu**, Mohan Li, Lei Yu, Ka-Ho Chow, Chengwei Qin, Xingyu Wu, Wenpeng Xing, Shuguang Xiong, Meng Han. *Hallucination Across the Reasoning Lifecycle: Interface Visibility, Causal Evidence, and Release Control in Large Reasoning Models*. Under review at **ACM Computing Surveys (CSUR)**. [arXiv]
2. **Zhe Yu**, Zixuan Wang, Peidong Wang, Hehai Lin, Ruochen Zhao, Chengwei Qin. *Beyond Corrected Memory: Execution Consistency in Multi-Agent Systems*. Under review at **ICLR 2027**. [arXiv]
3. **Zhe Yu**, Yi Liu, Ying Zhang, Simin Chen. *LatentTriad: Pre-Action Representation Probing for Malicious Agent Skill Detection*. Under review at **ICLR 2027**. [PDF]
4. **Zhe Yu**, Wenpeng Xing, Zhenhua Xu, Xingxing Yang, Meng Han. *Readable Before Actionable: Causal Tracing of Indirect Prompt Injection*. Under review at **ICLR 2027**. [arXiv]
5. **Zhe Yu**, Wenpeng Xing, Zhenhua Xu, Ruiqi Zhang, Meng Han. *Decodable but Unresolved: Chain-of-Thought Conflicts in Reasoning Models*. Under review at **ICLR 2027**. [PDF]
6. **Zhe Yu**, Yunzhao Wei, Wenpeng Xing, Wenjie Luo, Zaobo He, Quan Chen, Meng Han. *PLAIT: From Hallucination Scores to Parent-Preserving Audit Plans*. Under review at **ICLR 2027**. [PDF]
7. **Zhe Yu***, Wenpeng Xing*, Bo Yang, Chen Ye, Gaolei Li, Yunzhao Wei, Meng Han. *The Attribution Blind Spot: Detecting When Language Models Rely on Memory Rather Than Retrieved Context*. Under review at **ICLR 2027**. [arXiv]
8. **Zhe Yu***, Wenpeng Xing*, Chen Ye, Xuyang Teng, Bo Yang, Changting Lin, Meng Han. *Calibration Does Not Certify a Contrast: Identification and Correction for LLM Evaluators*. Under review at **AAAI 2027**. [PDF]
9. **Zhe Yu***, Wenpeng Xing*, Gaolei Li, Shuguang Xiong, Hongzhi Wang, Xuyang Teng, Meng Han. *Cordon-MAS: Defending RAG against Knowledge Poisoning via Information-Flow Control*. Under review at **ARR**. [arXiv]
10. Zhenhua Xu, Qichen Liu, Zhebo Wang, **Zhe Yu**, Xixiang Zhao, Wenpeng Xing, Dezhang Kong, Mohan Li, Meng Han. *Fingerprint Vector: Enabling Scalable and Efficient Model Fingerprint Transfer via Vector Addition*. Under review at **ARR (October 2026)**. [PDF]

* Equal contribution. Manuscript statuses updated October 12, 2026.

Patents

1. **Zhe Yu**, Wenpeng Xing, Meng Han. *A hallucination detection method based on dual-path internal state forcing logic for retrieval-augmented generation in large language models*. Chinese Patent Application No. 202610260408X; under review.
2. Meng Han, **Zhe Yu**, Jiayan Hu, Rongchang Li, Wenpeng Xing, Jingyi Yu, Zhen Hong, et al.. *A post-processing method, system, device, and medium for hallucination detection in large language models based on adaptive order statistics aggregation*. Chinese Patent Application No. 2026107898102; filed June 3, 2026; pending.
3. **Zhe Yu**, Jiayan Hu, Jingyi Yu, Weihang Yu, Wenpeng Xing, Jing Xiong, Yourong Chen, Zhen Hong, et al.. *A hallucination detection method, system, and device for large language models based on multi-dimensional heterogeneous feature fusion*. Chinese Patent Application No. 2026107899270; filed June 3, 2026; pending.